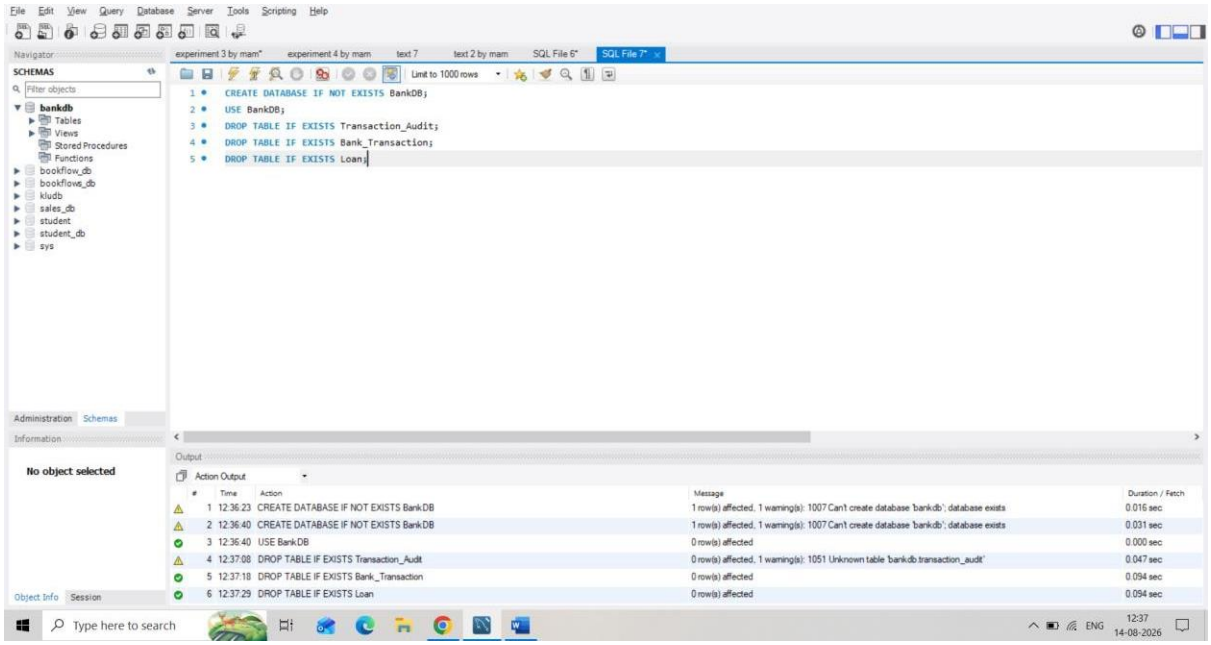
SKILL 4

TRIGGERS:




```
5 •  DROP TABLE IF EXISTS Loan;
6 •  DROP TABLE IF EXISTS Account;
7 •  DROP TABLE IF EXISTS Customer;
```

6	12:37:29	DROP TABLE IF EXISTS Loan	0 row(s) affected
7	12:37:51	DROP TABLE IF EXISTS Account	0 row(s) affected
8	12:38:01	DROP TABLE IF EXISTS Customer	0 row(s) affected

```sql
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2) DEFAULT 0,
    Branch VARCHAR(50),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
```

9	12:38:33	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	0 row(s) affected
10	12:38:47	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARC...	0 row(s) affected

```sql
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (Account_No) REFERENCES Account(Account_No)
);
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID)
);
```

11	12:39:15	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_...	0 row(s) affected
12	12:39:29	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected

```sql
CREATE TABLE Transaction_Audit (
    Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
    Transaction_ID INT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

13 12:39:57 CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID I... 0 row(s) affected

```sql
INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES
(10001, 101, 'Savings', 50000.00, 'Hyderabad'),
(10002, 102, 'Savings', 75000.00, 'Vijayawada'),
(10003, 103, 'Current', 120000.00, 'Bangalore'),
(10004, 104, 'Savings', 45000.00, 'Chennai'),
(10005, 105, 'Current', 90000.00, 'Hyderabad');
INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES
(10001, 'DEPOSIT', 10000.00),
(10002, 'DEPOSIT', 15000.00),
(10003, 'WITHDRAW', 20000.00),
(10004, 'DEPOSIT', 5000.00),
(10005, 'WITHDRAW', 10000.00);
INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES
(501, 101, 'Home Loan', 5000000.00, 7.50),
(502, 102, 'Education Loan', 1000000.00, 6.50),
(503, 103, 'Car Loan', 800000.00, 8.20),
(504, 104, 'Personal Loan', 500000.00, 10.50);
```

Output

Action Output

#	Time	Action	Message
1	13:33:17	USE BankDB	0 row(s) affected
2	13:33:22	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sa...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
3	13:33:36	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
4	13:33:48	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'Ho...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0

```sql
USE BankDB;
DELIMITER //

DROP PROCEDURE IF EXISTS GetAllCustomers //
CREATE PROCEDURE GetAllCustomers()
BEGIN
    SELECT * FROM Customer;
END //

DROP PROCEDURE IF EXISTS GetAccountDetails //
CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT)
BEGIN
    SELECT * FROM Account WHERE Account_No = p_Account_No;
END //

DROP PROCEDURE IF EXISTS GetCustomerAccounts //
CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT)
BEGIN
    SELECT C.Customer_ID, C.Customer_Name, A.Account_No, A.Account_Type, A.Balance, A.Branch
```

```
        FROM Customer C
        JOIN Account A ON C.Customer_ID = A.Customer_ID
        WHERE C.Customer_ID = p_Customer_ID;
    END //

    DROP PROCEDURE IF EXISTS DepositMoney //
    CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2))
    BEGIN
        UPDATE Account SET Balance = Balance + p_Amount WHERE Account_No = p_Account_No;
    END //

    DROP PROCEDURE IF EXISTS WithdrawMoney //
    CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2))
    BEGIN
        UPDATE Account SET Balance = Balance - p_Amount WHERE Account_No = p_Account_No;
    END //

    DROP PROCEDURE IF EXISTS TransferMoney //
    CREATE PROCEDURE TransferMoney(
        IN SenderAccount INT,
        IN ReceiverAccount INT,
```

```
        IN TransferAmount DECIMAL(12,2)
    )
    BEGIN
        DECLARE SenderBalance DECIMAL(12,2);
        SELECT Balance INTO SenderBalance FROM Account WHERE Account_No = SenderAccount;

        IF SenderBalance < TransferAmount THEN
            SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Transfer failed: Insufficient balance';
        ELSE
            UPDATE Account SET Balance = Balance - TransferAmount WHERE Account_No = SenderAccount;
            UPDATE Account SET Balance = Balance + TransferAmount WHERE Account_No = ReceiverAccount;
        END IF;
    END //

    DROP PROCEDURE IF EXISTS GetCustomerLoans //
    CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT)
    BEGIN
        SELECT C.Customer_Name, L.Loan_ID, L.Loan_Type, L.Loan_Amount, L.Interest_Rate
        FROM Customer C
        JOIN Loan L ON C.Customer_ID = L.Customer_ID
        WHERE C.Customer_ID = p_Customer_ID;
```

```
113         DECLARE SenderBalance DECIMAL(12,2);
114         SELECT Balance INTO SenderBalance FROM Account WHERE Account_No = SenderAccount;
115
116         IF SenderBalance < TransferAmount THEN
117             SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Transfer failed: Insufficient balance';
118         ELSE
119             UPDATE Account SET Balance = Balance - TransferAmount WHERE Account_No = SenderAccount;
120             UPDATE Account SET Balance = Balance + TransferAmount WHERE Account_No = ReceiverAccount;
121         END IF;
122     END //
123
124     DROP PROCEDURE IF EXISTS GetCustomerLoans //
125     CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT)
126     BEGIN
127         SELECT C.Customer_Name, L.Loan_ID, L.Loan_Type, L.Loan_Amount, L.Interest_Rate
128         FROM Customer C
129         JOIN Loan L ON C.Customer_ID = L.Customer_ID
130         WHERE C.Customer_ID = p_Customer_ID;
131     END //
132
133     DROP PROCEDURE IF EXISTS HighBalanceAccounts //
```

```
134     CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(12,2))
135     BEGIN
136         SELECT * FROM Account WHERE Balance >= MinimumBalance ORDER BY Balance DESC;
137     END //
138
139     DROP PROCEDURE IF EXISTS GetBalance //
140     CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMAL(12,2))
141     BEGIN
142         SELECT Balance INTO p_Balance FROM Account WHERE Account_No = p_Account_No;
143     END //
144
145     DELIMITER ;
```

✓	14	13:43:26	USE BankDB	0 row(s) affected
✓	15	13:43:26	DROP PROCEDURE IF EXISTS GetAllCustomers	0 row(s) affected
✓	16	13:43:26	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; END	0 row(s) affected
⚠	17	13:43:26	DROP PROCEDURE IF EXISTS GetAccountDetails	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.GetAccountDetails does not exist
✓	18	13:43:26	CREATE PROCEDURE GetAccountDetails(IN p_Account_No INT) BEGIN SELECT * FROM Account WHE...	0 row(s) affected
⚠	19	13:43:26	DROP PROCEDURE IF EXISTS GetCustomerAccounts	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.GetCustomerAccounts does not exist

⚠	19	13:43:26	DROP PROCEDURE IF EXISTS GetCustomerAccounts	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.GetCustomerAccounts does n
✓	20	13:43:26	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT C.Customer_ID, C.C...	0 row(s) affected
⚠	21	13:43:26	DROP PROCEDURE IF EXISTS DepositMoney	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.DepositMoney does not exist
✓	22	13:43:26	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPDA...	0 row(s) affected
⚠	23	13:43:26	DROP PROCEDURE IF EXISTS WithdrawMoney	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.WithdrawMoney does not exis
✓	24	13:43:26	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPD...	0 row(s) affected

⚠	23	13:43:26	DROP PROCEDURE IF EXISTS WithdrawMoney	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.WithdrawMoney does not exist
✓	24	13:43:26	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN UPD...	0 row(s) affected
⚠	25	13:43:26	DROP PROCEDURE IF EXISTS TransferMoney	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.TransferMoney does not exist
✓	26	13:43:26	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN ReceiverAccount INT, IN Transfer...	0 row(s) affected
⚠	27	13:43:26	DROP PROCEDURE IF EXISTS GetCustomerLoans	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.GetCustomerLoans does not exist
✓	28	13:43:26	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_Name, L.L...	0 row(s) affected

#	Time	Action	Message
27	13:43:26	DROP PROCEDURE IF EXISTS GetCustomerLoans	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.GetCustomerLoans does not exist
28	13:43:26	CREATE PROCEDURE GetCustomerLoans(IN p_Customer_ID INT) BEGIN SELECT C.Customer_Name, L.L...	0 row(s) affected
29	13:43:26	DROP PROCEDURE IF EXISTS HighBalanceAccounts	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.HighBalanceAccounts does not exist
30	13:43:26	CREATE PROCEDURE HighBalanceAccounts(IN MinimumBalance DECIMAL(12,2)) BEGIN SELECT * FRO...	0 row(s) affected
31	13:43:26	DROP PROCEDURE IF EXISTS GetBalance	0 row(s) affected, 1 warning(s): 1305 PROCEDURE bankdb.GetBalance does not exist
32	13:43:26	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMAL(12,2)) BEGIN SELE...	0 row(s) affected

```
DROP TRIGGER IF EXISTS CheckBalance //
CREATE TRIGGER CheckBalance
BEFORE UPDATE ON Account
FOR EACH ROW
BEGIN
    IF NEW.Balance < 0 THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';
    END IF;
END //

DROP TRIGGER IF EXISTS CheckTransactionAmount //
CREATE TRIGGER CheckTransactionAmount
BEFORE INSERT ON Bank_Transaction
```

```
FOR EACH ROW
BEGIN
    IF NEW.Amount <= 0 THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Transaction amount must be greater than zero';
    END IF;
END //

DROP TRIGGER IF EXISTS TransactionAudit //
CREATE TRIGGER TransactionAudit
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    INSERT INTO Transaction_Audit (Transaction_ID, Account_No, Transaction_Type, Amount)
    VALUES (NEW.Transaction_ID, NEW.Account_No, NEW.Transaction_Type, NEW.Amount);
END //

DROP TRIGGER IF EXISTS UpdateBalanceAfterTransaction //
CREATE TRIGGER UpdateBalanceAfterTransaction
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
```

```
    DECLARE CurrentBalance DECIMAL(12,2);
    SELECT Balance INTO CurrentBalance FROM Account WHERE Account_No = NEW.Account_No;

    IF NEW.Transaction_Type = 'WITHDRAW' AND NEW.Amount > CurrentBalance THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Withdrawal failed: Insufficient balance';
    END IF;
END //

DELIMITER ;
```

⚠	34	13:46:04	DROP TRIGGER IF EXISTS CheckBalance	0 row(s) affected, 1 warning(s): 1360 Trigger does not exist
✔	35	13:46:04	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Bala...	0 row(s) affected
⚠	36	13:46:04	DROP TRIGGER IF EXISTS CheckTransactionAmount	0 row(s) affected, 1 warning(s): 1360 Trigger does not exist
✔	37	13:46:04	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	0 row(s) affected
⚠	38	13:46:04	DROP TRIGGER IF EXISTS TransactionAudit	0 row(s) affected, 1 warning(s): 1360 Trigger does not exist

✔	39	13:46:04	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INS...	0 row(s) affected
⚠	40	13:46:04	DROP TRIGGER IF EXISTS UpdateBalanceAfterTransaction	0 row(s) affected, 1 warning(s): 1360 Trigger does not exist
✔	41	13:46:04	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Transaction FOR EACH ROW...	0 row(s) affected
⚠	42	13:46:04	DROP TRIGGER IF EXISTS PreventInsufficientWithdrawal	0 row(s) affected, 1 warning(s): 1360 Trigger does not exist

⚠	38	13:46:04	DROP TRIGGER IF EXISTS TransactionAudit	0 row(s) affected, 1 warning(s): 1360 Trigger does not exist
✔	39	13:46:04	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INS...	0 row(s) affected
⚠	40	13:46:04	DROP TRIGGER IF EXISTS UpdateBalanceAfterTransaction	0 row(s) affected, 1 warning(s): 1360 Trigger does not exist
✔	41	13:46:04	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Transaction FOR EACH ROW...	0 row(s) affected
⚠	42	13:46:04	DROP TRIGGER IF EXISTS PreventInsufficientWithdrawal	0 row(s) affected, 1 warning(s): 1360 Trigger does not exist
✔	43	13:46:04	CREATE TRIGGER PreventInsufficientWithdrawal BEFORE INSERT ON Bank_Transaction FOR EACH ROW ...	0 row(s) affected

```
USE BankDB;
DELIMITER //

DROP TRIGGER IF EXISTS CheckBalance //
CREATE TRIGGER CheckBalance
BEFORE UPDATE ON Account
FOR EACH ROW
BEGIN
    IF NEW.Balance < 0 THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';
    END IF;
END //

DROP TRIGGER IF EXISTS CheckTransactionAmount //
CREATE TRIGGER CheckTransactionAmount
BEFORE INSERT ON Bank_Transaction
```

✔	45	13:48:06	DROP TRIGGER IF EXISTS CheckBalance	0 row(s) affected
✔	46	13:48:06	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW BEGIN IF NEW.Bala...	0 row(s) affected
✔	47	13:48:06	DROP TRIGGER IF EXISTS CheckTransactionAmount	0 row(s) affected
✔	48	13:48:06	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	0 row(s) affected

```
FOR EACH ROW
BEGIN
    IF NEW.Amount <= 0 THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Transaction amount must be greater than zero';
    END IF;
END //

DROP TRIGGER IF EXISTS TransactionAudit //
CREATE TRIGGER TransactionAudit
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    INSERT INTO Transaction_Audit (Transaction_ID, Account_No, Transaction_Type, Amount)
    VALUES (NEW.Transaction_ID, NEW.Account_No, NEW.Transaction_Type, NEW.Amount);
END //

DROP TRIGGER IF EXISTS UpdateBalanceAfterTransaction //
CREATE TRIGGER UpdateBalanceAfterTransaction
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
```

49	13:48:06	DROP TRIGGER IF EXISTS TransactionAudit	0 row(s) affected
50	13:48:06	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INS...	0 row(s) affected
51	13:48:06	DROP TRIGGER IF EXISTS UpdateBalanceAfterTransaction	0 row(s) affected
52	13:48:06	CREATE TRIGGER UpdateBalanceAfterTransaction AFTER INSERT ON Bank_Transaction FOR EACH ROW...	0 row(s) affected
53	13:48:06	DROP TRIGGER IF EXISTS PreventInsufficientWithdrawal	0 row(s) affected
54	13:48:06	CREATE TRIGGER PreventInsufficientWithdrawal BEFORE INSERT ON Bank_Transaction FOR EACH ROW	0 row(s) affected

```
    DECLARE CurrentBalance DECIMAL(12,2);
    SELECT Balance INTO CurrentBalance FROM Account WHERE Account_No = NEW.Account_No;

    IF NEW.Transaction_Type = 'WITHDRAW' AND NEW.Amount > CurrentBalance THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Withdrawal failed: Insufficient balance';
    END IF;
END //

DELIMITER ;
```

```
CALL GetAllCustomers();
CALL GetAccountDetails(10001);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

```
CALL GetAccountDetails(10001);
CALL GetCustomerAccounts(101);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

```
CALL GetCustomerAccounts(101);
CALL HighBalanceAccounts(50000);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	10001	Savings	50000.00	Hyderabad

```
CALL HighBalanceAccounts(50000);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10001	101	Savings	50000.00	Hyderabad

```
SELECT * FROM Transaction_Audit;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Audit_ID	Transaction_ID	Account_No	Transaction_Type	Amount	Audit_Date
NULL	NULL	NULL	NULL	NULL	NULL

SKILL 3

```sql
CREATE DATABASE BankDB;
```

1 15:18:02 CREATE DATABASE BankDB 1 row(s) affected

```sql
USE BankDB;
```

2 15:18:19 USE BankDB

```sql
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    Email VARCHAR(100),
    City VARCHAR(50)
);
```

3 15:18:33 CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT N...

```sql
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

4 15:18:50 CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARC... 0 row(s) affected

```sql
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

5 15:19:06 CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account... 0 row(s) affected

```sql
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

6 15:19:20 CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ... 0 row(s) affected

```sql
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'Ravi Kumar', '9876543210', 'ravi@gmail.com', 'Hyderabad'),
(102, 'Priya Sharma', '9876543211', 'priya@gmail.com', 'Vijayawada'),
(103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
(104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
(105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad'),
(106, 'Anil Kumar', '9876543215', 'anil@gmail.com', 'Delhi'),
(107, 'Meena Reddy', '9876543216', 'meena@gmail.com', 'Mumbai'),
(108, 'Rahul Sharma', '9876543217', 'rahul@gmail.com', 'Pune'),
(109, 'Lakshmi Devi', '9876543218', 'lakshmi@gmail.com', 'Hyderabad'),
(110, 'Suresh Babu', '9876543219', 'suresh@gmail.com', 'Vijayawada');
```

7 15:19:38 INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Ravi Kumar', '987... 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

```sql
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad'),
(10006, 106, 'Savings', 65000, 'Delhi'),
(10007, 107, 'Current', 150000, 'Mumbai'),
(10008, 108, 'Savings', 35000, 'Pune'),
(10009, 109, 'Savings', 85000, 'Hyderabad'),
(10010, 110, 'Current', 110000, 'Vijayawada');
```

8 15:20:02 INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, '... 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

```sql
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10001, 'WITHDRAW', 5000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000),
(10006, 'DEPOSIT', 12000),
(10007, 'DEPOSIT', 25000),
(10008, 'WITHDRAW', 5000),
(10009, 'DEPOSIT', 20000),
(10010, 'WITHDRAW', 15000);
```

9 15:26:36 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10... 11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0

```sql
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),
(502, 102, 'Education Loan', 1000000, 6.5),
(503, 103, 'Car Loan', 800000, 8.2),
(504, 104, 'Personal Loan', 500000, 10.5),
(505, 105, 'Home Loan', 4000000, 7.2),
(506, 106, 'Car Loan', 900000, 8.5),
(507, 107, 'Business Loan', 3000000, 9.0),
(508, 109, 'Personal Loan', 600000, 10.0);
```

10 15:26:52 INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, '... 8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0

```sql
SELECT * FROM Customer;
SELECT * FROM Account;
SELECT * FROM Bank_Transaction;
SELECT * FROM Loan;
```

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hydera Chennai
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad
110	Suresh Babu	9876543219	suresh@gmail.com	Vijayawada
NULL	NULL	NULL	NULL	NULL

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-12 15:26:36
2	10001	WITHDRAW	5000.00	2026-08-12 15:26:36
3	10002	DEPOSIT	15000.00	2026-08-12 15:26:36
4	10003	WITHDRAW	20000.00	2026-08-12 15:26:36
5	10004	DEPOSIT	5000.00	2026-08-12 15:26:36
6	10005	WITHDRAW	10000.00	2026-08-12 15:26:36
7	10006	DEPOSIT	12000.00	2026-08-12 15:26:36
8	10007	DEPOSIT	25000.00	2026-08-12 15:26:36
9	10008	WITHDRAW	5000.00	2026-08-12 15:26:36
10	10009	DEPOSIT	20000.00	2026-08-12 15:26:36
11	10010	WITHDRAW	15000.00	2026-08-12 15:26:36
NULL	NULL	NULL	NULL	NULL

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20
506	106	Car Loan	900000.00	8.50
507	107	Business Loan	3000000.00	9.00
508	109	Personal Loan	600000.00	10.00
NULL	NULL	NULL	NULL	NULL

7.20

```sql
CREATE VIEW Customer_View AS
SELECT *
FROM Customer;
```

20 15:40:23 CREATE VIEW Customer_View AS SELECT * FROM Customer 0 row(s) affected

```sql
SELECT * FROM Customer_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad
110	Suresh Babu	9876543219	suresh@gmail.com	Vijayawada

```sql
CREATE VIEW Customer_Basic_View AS
SELECT
    Customer_ID,
    Customer_Name,
    City
FROM Customer;
```

22 15:41:33 CREATE VIEW Customer_Basic_View AS SELECT Customer_ID, Customer_Name, City FROM Customer 0 row(s) affected

```
105 •  SELECT * FROM Customer_Basic_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City
101	Ravi Kumar	Hyderabad
102	Priya Sharma	Vijayawada
103	Arjun Reddy	Bangalore
104	Sneha Rao	Chennai
105	Kiran Kumar	Hyderabad
106	Anil Kumar	Delhi
107	Meena Reddy	Mumbai
108	Rahul Sharma	Pune
109	Lakshmi Devi	Hyderabad
110	Suresh Babu	Vijayawada

```
CREATE VIEW Account_View AS
SELECT
    Account_No,
    Account_Type,
    Balance,
    Branch
FROM Account;
```

24 15:42:45 CREATE VIEW Account_View AS SELECT Account_No, Account_Type, Balance, Branch FROM ... 0 row(s) affected

```
113 •  SELECT * FROM Account_View;
```

Result Grid | Filter Rows: | Export:

Account_No	Account_Type	Balance	Branch
10001	Savings	50000.00	Hyderabad
10002	Savings	75000.00	Vijayawada
10003	Current	120000.00	Bangalore
10004	Savings	45000.00	Chennai
10005	Current	90000.00	Hyderabad
10006	Savings	65000.00	Delhi
10007	Current	150000.00	Mumbai
10008	Savings	35000.00	Pune
10009	Savings	85000.00	Hyderabad
10010	Current	110000.00	Vijayawada

```
CREATE VIEW Savings_Account_View AS
SELECT *
FROM Account
WHERE Account_Type = 'Savings';
```

26 15:43:59 CREATE VIEW Savings_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Savings' 0 row(s) affected

```
118 •  SELECT * FROM Savings_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap C

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10004	104	Savings	45000.00	Chennai
10006	106	Savings	65000.00	Delhi
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad

```
CREATE VIEW Current_Account_View AS
SELECT *
FROM Account
WHERE Account_Type = 'Current';
```

28 15:45:06 CREATE VIEW Current_Account_View AS SELECT * FROM Account WHERE Account_Type = 'Current' 0 row(s) affected

```
123 •  SELECT * FROM Current_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10007	107	Current	150000.00	Mumbai
10010	110	Current	150000.00	Vijayawada

```
CREATE VIEW High_Balance_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance
FROM Account
WHERE Balance > 100000;
```

30 15:46:24 CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, Account_Type, Balan... 0 row(s) affected

```
132 •  SELECT * FROM High_Balance_View;
```

Result Grid | Filter Rows: | Export:

Account_No	Customer_ID	Account_Type	Balance
10003	103	Current	120000.00
10007	107	Current	150000.00
10010	110	Current	110000.00

```
CREATE VIEW Hyderabad_Account_View AS
SELECT *
FROM Account
WHERE Branch = 'Hyderabad';
```

3 21:24:25 CREATE VIEW Hyderabad_Account_View AS SELECT * FROM Account WHERE Branch = 'Hyderabad' 0 row(s) affected

```
137 •  SELECT * FROM Hyderabad_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10005	105	Current	90000.00	Hyderabad
10009	109	Savings	85000.00	Hyderabad

```
CREATE VIEW Low_Balance_View AS
SELECT
    Account_No,
    Customer_ID,
    Balance
FROM Account
WHERE Balance < 50000;
```

5 21:25:39 CREATE VIEW Low_Balance_View AS SELECT Account_No, Customer_ID, Balance FROM Account ... 0 row(s) affected

```
145 •  SELECT * FROM Low_Balance_View;
```

Result Grid | Filter Rows:

Account_No	Customer_ID	Balance
10004	104	45000.00
10008	108	35000.00

```sql
CREATE VIEW Customer_Account_View AS
SELECT
    C.Customer_ID,
    C.Customer_Name,
    C.City,
    A.Account_No,
    A.Account_Type,
    A.Balance,
    A.Branch
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID;
```

7 21:26:40 CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A.... 0 row(s) affected

```sql
158 • SELECT * FROM Customer_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	Hyderabad	10001	Savings	50000.00	Hyderabad
102	Priya Sharma	Vijayawada	10002	Savings	75000.00	Vijayawada
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad
106	Anil Kumar	Delhi	10006	Savings	65000.00	Delhi
107	Meena Reddy	Mumbai	10007	Current	150000.00	Mumbai
108	Rahul Sharma	Pune	10008	Savings	35000.00	Pune
109	Lakshmi Devi	Hyderabad	10009	Savings	85000.00	Hyderabad
110	Suresh Babu	Vijayawada	10010	Current	110000.00	Vijayawada

Delhi

```sql
CREATE VIEW Hyderabad_Customer_Accounts AS
SELECT
    C.Customer_Name,
    C.City,
    A.Account_No,
    A.Account_Type,
    A.Balance
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID
WHERE A.Branch = 'Hyderabad';
```

9 21:27:40 CREATE VIEW Hyderabad_Customer_Accounts AS SELECT C.Customer_Name, C.City, A.Account_N... 0 row(s) affected

```sql
170 • SELECT * FROM Hyderabad_Customer_Accounts;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_Name	City	Account_No	Account_Type	Balance
Ravi Kumar	Hyderabad	10001	Savings	50000.00
Kiran Kumar	Hyderabad	10005	Current	90000.00
Lakshmi Devi	Hyderabad	10009	Savings	85000.00

```sql
CREATE VIEW Customer_Loan_View AS
SELECT
    C.Customer_ID,
    C.Customer_Name,
    C.City,
    L.Loan_ID,
    L.Loan_Type,
    L.Loan_Amount,
    L.Interest_Rate
FROM Customer C
JOIN Loan L
ON C.Customer_ID = L.Customer_ID;
```

11 21:28:46 CREATE VIEW Customer_Loan_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, L.Lo... 0 row(s) affected

```
183 •  SELECT * FROM Customer_Loan_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
101	Ravi Kumar	Hyderabad	501	Home Loan	5000000.00	7.50
102	Priya Sharma	Vijayawada	502	Education Loan	1000000.00	6.50
103	Arjun Reddy	Bangalore	503	Car Loan	800000.00	8.20
104	Sneha Rao	Chennai	504	Personal Loan	500000.00	10.50
105	Kiran Kumar	Hyderabad	505	Home Loan	4000000.00	7.20
106	Anil Kumar	Delhi	506	Car Loan	900000.00	8.50
107	Meena Reddy	Mumbai	507	Business Loan	3000000.00	9.00
109	Lakshmi Devi	Hyderabad	508	Personal Loan	600000.00	10.00

```
CREATE VIEW Customer_Banking_View AS
SELECT
    C.Customer_ID,
    C.Customer_Name,
    C.City,
    A.Account_No,
    A.Account_Type,
    A.Balance,
    A.Branch,
    L.Loan_Type,
    L.Loan_Amount
FROM Customer C
LEFT JOIN Account A
ON C.Customer_ID = A.Customer_ID
LEFT JOIN Loan L
ON C.Customer_ID = L.Customer_ID;
```

13 21:29:48 CREATE VIEW Customer_Banking_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A.... 0 row(s) affected

```
200 •  SELECT * FROM Customer_Banking_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch	Loan_Type	Loan_Amount
101	Ravi Kumar	Hyderabad	10001	Savings	50000.00	Hyderabad	Home Loan	5000000.00
102	Priya Sharma	Vijayawada	10002	Savings	75000.00	Vijayawada	Education Loan	1000000.00
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore	Car Loan	800000.00
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai	Personal Loan	500000.00
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad	Home Loan	4000000.00
106	Anil Kumar	Delhi	10006	Savings	65000.00	Delhi	Car Loan	900000.00
107	Meena Reddy	Mumbai	10007	Current	150000.00	Mumbai	Business Loan	3000000.00
108	Rahul Sharma	Pune	10008	Savings	35000.0	35000.00	NULL	NULL
109	Lakshmi Devi	Hyderabad	10009	Savings	85000.00	Hyderabad	Personal Loan	600000.00
110	Suresh Babu	Vijayawada	10010	Current	110000.00	Vijayawada	NULL	NULL

```
CREATE VIEW Total_Bank_Balance AS
SELECT
    SUM(Balance) AS Total_Balance
FROM Account;
```

15 21:30:54 CREATE VIEW Total_Bank_Balance AS SELECT SUM(Balance) AS Total_Balance FROM Account

```
205 •  SELECT * FROM Total_Bank_Balance;
```

Result Grid | Filter Rows:

Total_Balance
825000.00

```
CREATE VIEW Average_Account_Balance AS
SELECT
    AVG(Balance) AS Average_Balance
FROM Account;
```

17 21:31:44 CREATE VIEW Average_Account_Balance AS SELECT AVG(Balance) AS Average_Balance FROM Acco... 0 row(s) affected

```
210 •  SELECT * FROM Average_Account_Balance;
```

Result Grid | Filter Rows: | Export:

Average_Balance
82500.000000

```sql
CREATE VIEW Maximum_Balance_View AS
SELECT
    MAX(Balance) AS Maximum_Balance
FROM Account;
```

19 21:32:39 CREATE VIEW Maximum_Balance_View AS SELECT MAX(Balance) AS Maximum_Balance FROM Account 0 row(s) affected

```sql
SELECT * FROM Maximum_Balance_View;
```

Result Grid | Filter Rows: | Expo

Maximum_Balance
150000.00

```sql
CREATE VIEW Minimum_Balance_View AS
SELECT
    MIN(Balance) AS Minimum_Balance
FROM Account;
```

21 21:33:29 CREATE VIEW Minimum_Balance_View AS SELECT MIN(Balance) AS Minimum_Balance FROM Account 0 row(s) affected

```sql
SELECT * FROM Minimum_Balance_View;
```

Result Grid | Filter Rows: | Exp

Minimum_Balance
35000.00

```sql
CREATE VIEW Account_Count_View AS
SELECT
    COUNT(*) AS Total_Accounts
FROM Account;
```

23 21:35:06 CREATE VIEW Account_Count_View AS SELECT COUNT(*) AS Total_Accounts FROM Account

```
SELECT * FROM Account_Count_View;
```

Result Grid | Filter Rows: | Ex

Total_Accounts
10

```
CREATE VIEW Branch_Account_Count AS
SELECT
    Branch,
    COUNT(*) AS Number_of_Accounts
FROM Account
GROUP BY Branch;
```

25 21:36:05 CREATE VIEW Branch_Account_Count AS SELECT Branch, COUNT(*) AS Number_of_Accounts FRO... 0 row(s) affected

```
SELECT * FROM Branch_Account_Count;
```

Result Grid | Filter Rows: | Export:

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2
Bangalore	1
Chennai	1
Delhi	1
Mumbai	1
Pune	1

```
CREATE VIEW Account_Type_Count AS
SELECT
    Account_Type,
    COUNT(*) AS Number_of_Accounts
FROM Account
GROUP BY Account_Type;
```

29 21:38:30 CREATE VIEW Account_Type_Count AS SELECT Account_Type, COUNT(*) AS Number_of_Accounts ... 0 row(s) affected

```
246 •  SELECT * FROM Account_Type_Count;
```

Result Grid | Filter Rows: | Expo

Account_Type	Number_of_Accounts
Savings	6
Current	4

```
CREATE VIEW Account_Type_Balance AS
SELECT
    Account_Type,
    SUM(Balance) AS Total_Balance
FROM Account
GROUP BY Account_Type;
```

31 21:39:24 CREATE VIEW Account_Type_Balance AS SELECT Account_Type, SUM(Balance) AS Total_Balance ... 0 row(s) affected

```
253 •  SELECT * FROM Account_Type_Balance;
```

Result Grid | Filter Rows: | Expo

Account_Type	Total_Balance
Savings	355000.00
Current	470000.00

```
•  CREATE VIEW Multiple_Account_Branches AS
   SELECT
       Branch,
       COUNT(*) AS Number_of_Accounts
   FROM Account
   GROUP BY Branch
   HAVING COUNT(*) > 1;
```

33 21:40:22 CREATE VIEW Multiple_Account_Branches AS SELECT Branch, COUNT(*) AS Number_of_Accounts F... 0 row(s) affected

```
261 •  SELECT * FROM Multiple_Account_Branches;
```

Result Grid | Filter Rows: | Export:

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2

```
CREATE VIEW Rich_Branches AS
SELECT
    Branch,
    SUM(Balance) AS Total_Balance
FROM Account
GROUP BY Branch
HAVING SUM(Balance) > 100000;
```

35 21:41:17 CREATE VIEW Rich_Branches AS SELECT Branch, SUM(Balance) AS Total_Balance FROM Account ... 0 row(s) affected

```
269 •  SELECT * FROM Rich_Branches;
```

Result Grid | Filter Rows:

Branch	Total_Balance
Hyderabad	225000.00
Vijayawada	185000.00
Bangalore	120000.00
Mumbai	150000.00

```
CREATE VIEW Deposit_Transaction_View AS
SELECT *
FROM Bank_Transaction
WHERE Transaction_Type = 'DEPOSIT';
```

37 21:42:16 CREATE VIEW Deposit_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_Ty... 0 row(s) affected

```
274 •  SELECT * FROM Deposit_Transaction_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-12 15:26:36
3	10002	DEPOSIT	15000.00	2026-08-12 15:26:36
5	10004	DEPOSIT	5000.00	2026-08-12 15:26:36
7	10006	DEPOSIT	12000.00	2026-08-12 15:26:36
8	10007	DEPOSIT	25000.00	2026-08-12 15:26:36
10	10009	DEPOSIT	20000.00	2026-08-12 15:26:36

```
CREATE VIEW Withdrawal_Transaction_View AS
SELECT *
FROM Bank_Transaction
WHERE Transaction_Type = 'WITHDRAW';
```

39 21:43:19 CREATE VIEW Withdrawal_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_... 0 row(s) affected

```
279 •  SELECT * FROM Withdrawal_Transaction_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
2	10001	WITHDRAW	5000.00	2026-08-12 15:26:36
4	10003	WITHDRAW	20000.00	2026-08-12 15:26:36
6	10005	WITHDRAW	10000.00	2026-08-12 15:26:36
9	10008	WITHDRAW	5000.00	2026-08-12 15:26:36
11	10010	WITHDRAW	15000.00	2026-08-12 15:26:36

```sql
CREATE VIEW Customer_Transaction_View AS
SELECT
    C.Customer_Name,
    A.Account_No,
    A.Account_Type,
    T.Transaction_ID,
    T.Transaction_Type,
    T.Amount,
    T.Transaction_Date
FROM Customer C
JOIN Account A
ON C.Customer_ID = A.Customer_ID
JOIN Bank_Transaction T
ON A.Account_No = T.Account_No;
```

41 21:44:13 CREATE VIEW Customer_Transaction_View AS SELECT C.Customer_Name, A.Account_No, A.Acco... 0 row(s) affected

```sql
294 • SELECT * FROM Customer_Transaction_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_Name	Account_No	Account_Type	Transaction_ID	Transaction_Type	Amount	Transaction_Date
Ravi Kumar	10001	Savings	1	DEPOSIT	10000.00	2026-08-12 15:26:36
Ravi Kumar	10001	Savings	2	WITHDRAW	5000.00	2026-08-12 15:26:36
Priya Sharma	10002	Savings	3	DEPOSIT	15000.00	2026-08-12 15:26:36
Arjun Reddy	10003	Current	4	WITHDRAW	20000.00	2026-08-12 15:26:36
Sneha Rao	10004	Savings	5	DEPOSIT	5000.00	2026-08-12 15:26:36
Kiran Kumar	10005	Current	6	WITHDRAW	10000.00	2026-08-12 15:26:36
Anil Kumar	10006	Savings	7	DEPOSIT	12000.00	2026-08-12 15:26:36
Meena Reddy	10007	Current	8	DEPOSIT	25000.00	2026-08-12 15:26:36
Rahul Sharma	10008	Savings	9	WITHDRAW	5000.00	2026-08-12 15:26:36
Lakshmi Devi	10009	Savings	10	DEPOSIT	20000.00	2026-08-12 15:26:36
Suresh Babu	10010	Current	11	WITHDRAW	15000.00	2026-08-12 15:26:36

```sql
CREATE VIEW High_Value_Transaction_View AS
SELECT *
FROM Bank_Transaction
WHERE Amount > 10000;
```

43 21:45:09 CREATE VIEW High_Value_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Amount > 10... 0 row(s) affected

```
299 •  SELECT * FROM High_Value_Transaction_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
3	10002	DEPOSIT	15000.00	2026-08-12 15:26:36
4	10003	WITHDRAW	20000.00	2026-08-12 15:26:36
7	10006	DEPOSIT	12000.00	2026-08-12 15:26:36
8	10007	DEPOSIT	25000.00	2026-08-12 15:26:36
10	10009	DEPOSIT	20000.00	2026-08-12 15:26:36
11	10010	WITHDRAW	15000.00	2026-08-12 15:26:36

```
CREATE VIEW Balance_Ranking_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance
FROM Account
ORDER BY Balance DESC;
```

45 21:46:01 CREATE VIEW Balance_Ranking_View AS SELECT Account_No, Customer_ID, Account_Type, B... 0 row(s) affected

```
308 •  SELECT * FROM Balance_Ranking_View;
```

Result Grid | Filter Rows: | Export:

Account_No	Customer_ID	Account_Type	Balance
10007	107	Current	150000.00
10003	103	Current	120000.00
10010	110	Current	110000.00
10005	105	Current	90000.00
10009	109	Savings	85000.00
10002	102	Savings	75000.00
10006	106	Savings	65000.00
10001	101	Savings	50000.00
10004	104	Savings	45000.00
10008	108	Savings	35000.00

```sql
CREATE VIEW Customer_Name_View AS
SELECT
    Customer_ID,
    Customer_Name,
    City
FROM Customer
ORDER BY Customer_Name;
```

47 21:46:47 CREATE VIEW Customer_Name_View AS SELECT Customer_ID, Customer_Name, City FROM Custo... 0 row(s) affected

```sql
316 • SELECT * FROM Customer_Name_View;
```

Result Grid | Filter Rows:

Customer_ID	Customer_Name	City
106	Anil Kumar	Delhi
103	Arjun Reddy	Bangalore
105	Kiran Kumar	Hyderabad
109	Lakshmi Devi	Hyderabad
107	Meena Reddy	Mumbai
102	Priya Sharma	Vijayawada
108	Rahul Sharma	Pune
101	Ravi Kumar	Hyderabad
104	Sneha Rao	Chennai
110	Suresh Babu	Vijayawada

Pune

```sql
CREATE VIEW Loan_Interest_View AS
SELECT
    Loan_ID,
    Customer_ID,
    Loan_Type,
    Loan_Amount,
    Interest_Rate,
    (Loan_Amount * Interest_Rate / 100) AS Annual_Interest
FROM Loan;
```

49 21:47:38 CREATE VIEW Loan_Interest_View AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_Amoun... 0 row(s) affected

```
326 •  SELECT * FROM Loan_Interest_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest
501	101	Home Loan	5000000.00	7.50	375000.00000000
502	102	Education Loan	1000000.00	6.50	65000.00000000
503	103	Car Loan	800000.00	8.20	65600.00000000
504	104	Personal Loan	500000.00	10.50	52500.00000000
505	105	Home Loan	4000000.00	7.20	288000.00000000
506	106	Car Loan	900000.00	8.50	76500.00000000
507	107	Business Loan	3000000.00	9.00	270000.00000000
508	109	Personal Loan	600000.00	10.00	60000.00000000

```
CREATE VIEW Loan_Total_Amount_View AS
SELECT
    Loan_ID,
    Customer_ID,
    Loan_Type,
    Loan_Amount,
    Interest_Rate,
    Loan_Amount + (Loan_Amount * Interest_Rate / 100) AS Total_Amount
FROM Loan;
```

51 21:48:39 CREATE VIEW Loan_Total_Amount_View AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_... 0 row(s) affected

```
336 •  SELECT * FROM Loan_Total_Amount_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Total_Amount
501	101	Home Loan	5000000.00	7.50	5375000.00000000
502	102	Education Loan	1000000.00	6.50	1065000.00000000
503	103	Car Loan	800000.00	8.20	865600.00000000
504	104	Personal Loan	500000.00	10.50	552500.00000000
505	105	Home Loan	4000000.00	7.20	4288000.00000000
506	106	Car Loan	900000.00	8.50	976500.00000000
507	107	Business Loan	3000000.00	9.00	3270000.00000000
508	109	Personal Loan	600000.00	10.00	660000.00000000

```
SELECT *
FROM High_Balance_View
WHERE Balance > 120000;
```

Result Grid | Filter Rows: | Export:

Account_No	Customer_ID	Account_Type	Balance
10007	107	Current	150000.00

```
SELECT *
FROM Savings_Account_View
WHERE Balance > 60000;
```

Result Grid | Filter Rows: | Export: | Wrap C

Account_No	Customer_ID	Account_Type	Balance	Branch
10002	102	Savings	75000.00	Vijayawada
10006	106	Savings	65000.00	Delhi
10009	109	Savings	85000.00	Hyderabad

```
SELECT *
FROM Hyderabad_Account_View
WHERE Balance > 50000;
```

Result Grid | Filter Rows: | Export: | Wrap Cel

Account_No	Customer_ID	Account_Type	Balance	Branch
10005	105	Current	90000.00	Hyderabad
10009	109	Savings	85000.00	Hyderabad

```
346 •  SELECT *
347    FROM Customer_Basic_View
348    WHERE City = 'Hyderabad';
```

Result Grid | Filter Rows: | Ex

Customer_ID	Customer_Name	City
101	Ravi Kumar	Hyderabad
105	Kiran Kumar	Hyderabad
109	Lakshmi Devi	Hyderabad

```
349 •  SELECT *
350    FROM Customer_Account_View
351    ORDER BY Balance DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
107	Meena Reddy	Mumbai	10007	Current	150000.00	Mumbai
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore
110	Suresh Babu	Vijayawada	10010	Current	110000.00	Vijayawada
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad
109	Lakshmi Devi	Hyderabad	10009	Savings	85000.00	Hyderabad
102	Priya Sharma	Vijayawada	10002	Savings	75000.00	Vijayawada
106	Anil Kumar	Delhi	10006	Savings	65000.00	Delhi
101	Ravi Kumar	Hyderabad	10001	Savings	50000.00	Hyderabad
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai
108	Rahul Sharma	Pune	10008	Savings	35000.00	Pune

Current

```
352 •  SELECT *
353    FROM Customer_Loan_View
354    WHERE Loan_Amount > 1000000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
101	Ravi Kumar	Hyderabad	501	Home Loan	5000000.00	7.50
105	Kiran Kumar	Hyderabad	505	Home Loan	4000000.00	7.20
107	Meena Reddy	Mumbai	507	Business Loan	3000000.00	9.00

```
UPDATE Account_View
SET Balance = 60000
WHERE Account_No = 10001;
```

59 21:52:18 UPDATE Account_View SET Balance = 60000 WHERE Account_No = 10001 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

```sql
SELECT *
FROM Account
WHERE Account_No = 10001;
```

Result Grid | Filter Rows: | Edit:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

```sql
CREATE VIEW Simple_Account_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance,
    Branch
FROM Account;

INSERT INTO Simple_Account_View
VALUES
(10011, 101, 'Savings', 55000, 'Hyderabad');
```

62 21:53:35 INSERT INTO Simple_Account_View VALUES (10011, 101, 'Savings', 55000, 'Hyderabad') 1 row(s) affected

```
SELECT *
FROM Account;
```

Result Grid | Filter Rows: | Edit:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada
10011	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

```
DELETE FROM Simple_Account_View
WHERE Account_No = 10011;
```

64 21:54:28 DELETE FROM Simple_Account_View WHERE Account_No = 10011 1 row(s) affected

```
SELECT *
FROM Account;
```

Result Grid | Filter Rows: | Edit:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	65000.00	Delhi
10007	107	Current	150000.00	Mumbai
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
10010	110	Current	110000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

```
379 •  SHOW FULL TABLES
380    WHERE TABLE_TYPE = 'VIEW';
```

Result Grid | Filter Rows:

Tables_in_bankdb	Table_type
account_count_view	VIEW
account_type_balance	VIEW
account_type_count	VIEW
account_view	VIEW
average_account_balance	VIEW
balance_ranking_view	VIEW
branch_account_count	VIEW
branch_total_balance	VIEW
current_account_view	VIEW
customer_account_view	VIEW
customer_banking_view	VIEW
customer_basic_view	VIEW

```
381 •  SHOW CREATE VIEW Customer_Account_View;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

View	Create View	character_set_client	collation_connection
customer_account_view	CREATE ALGORITHM=UNDEFINED DEFINER=`r...	utf8mb4	utf8mb4_0900_ai_ci

```
382 •  DESCRIBE Customer_Account_View;
```

Result Grid | Filter Rows: | Export: | W

Field	Type	Null	Key	Default	Ex
Customer_ID	int	NO		NULL	
Customer_Name	varchar(100)	NO		NULL	
City	varchar(50)	YES		NULL	
Account_No	int	NO		NULL	
Account_Type	varchar(20)	YES		NULL	
Balance	decimal(12,2)	YES		NULL	
Branch	varchar(50)	YES		NULL	

```
383 •  DROP VIEW Customer_View;
```

69 21:56:46 DROP VIEW Customer_View

```
DROP VIEW
Customer_Basic_View,
Savings_Account_View,
Current_Account_View;
```

70	21:57:12	DROP VIEW Customer_Basic_View, Savings_Account_View, Current_Account_View	0 row(s) affected